Chapter 2: System Development Life Cycle (SDLC)

A system is a collection of components that interact to achieve a specific goal. In ICT, the System Development Life Cycle (SDLC) provides a structured framework for building high-quality information systems.

2.1 The Concept of a System

Basic elements of any system include:

- **Input:** Data or resources entering the system.
- **Process:** The conversion of input into output.
- **Output:** The finished product or information.

2.2 Manual vs. Computer-Based Systems

Feature	Manual System	Computer-Based System
Speed	Slow	Very Fast
Accuracy	Prone to human error	High accuracy
Storage	Physical files (bulky)	Digital storage (compact)
Efficiency	Lower	Higher

2.3 Stages of SDLC

1. **Identification of Requirements:** Understanding what the user needs and identifying the problems with the current system.

2. **Analysis:** Detailed study of the requirements to determine feasibility.
3. **Design:** Creating the blueprint, including user interfaces, databases, and system architecture.
4. **Development / Implementation:** Writing the actual code or setting up the physical system.
5. **Testing:** Checking for errors (debugging) to ensure the system works as expected.
6. **Deployment:** Releasing the system for actual use.
7. **Maintenance:** Updating and fixing the system over time.

2.4 SDLC Models

- **Waterfall Model:** A linear, sequential approach where each phase must be completed before the next begins. Best for projects with clear, unchanging requirements.
- **Iterative Incremental Model:** Develops the system in small, manageable parts or "increments," repeating cycles to improve the product.
- **Prototyping Model:** Creating a "mock-up" or experimental version of the system to get user feedback early.
- **Spiral Model:** Focuses on risk analysis, combining iterative development with systematic aspects of the waterfall model.